

2 The reaction between an acid and an alkali is exothermic. You will carry out a neutralisation experiment to determine the enthalpy change involved.

You will mix different volumes of an acid with a fixed volume of an alkali and measure the temperature rises that occur.

FB 4 is aqueous sodium hydroxide, NaOH.
FB 5 is 2.00 mol dm⁻³ hydrochloric acid, HCl.

(a) Method

- Use the thermometer to measure the initial temperature of **FB 4**. Record this initial temperature in the space for results.
- Support the cup in the 250 cm³ beaker.
- Fill one burette with **FB 5**. Label the burette **FB 5**.
- Fill the other burette with distilled water.

Experiment 1

- Use the 10 cm³ pipette to transfer 10.0 cm³ of **FB 4** into the cup.
- Add 9.00 cm³ of distilled water from the burette to the same cup.
- Add 1.00 cm³ of **FB 5** from the other burette to the same cup.
- Stir the mixture and use the thermometer to measure the maximum temperature. If necessary, tilt the cup so that the solution covers the bulb of the thermometer.
- Record the maximum temperature in Table 2.1.
- Empty, rinse and dry the cup ready for use in further experiments.

Further experiments

Repeat this method for Experiments 2–5, using 10.0 cm³ of **FB 4** and the volumes of water and **FB 5** shown in Table 2.1. In each case, measure and record the maximum temperature.

Carry out **two** further experiments, Experiments 6 and 7, which will enable you to determine more precisely the volume of **FB 5** that gives the largest maximum temperature. Record your measurements in Table 2.1.

Results

initial temperature of **FB 4** = °C

Table 2.1

experiment	volume of water /cm ³	volume of FB 5 /cm ³	maximum temperature/°C
1	9.00	1.00	
2	7.00	3.00	
3	5.00	5.00	
4	3.00	7.00	
5	1.00	9.00	
6			
7			

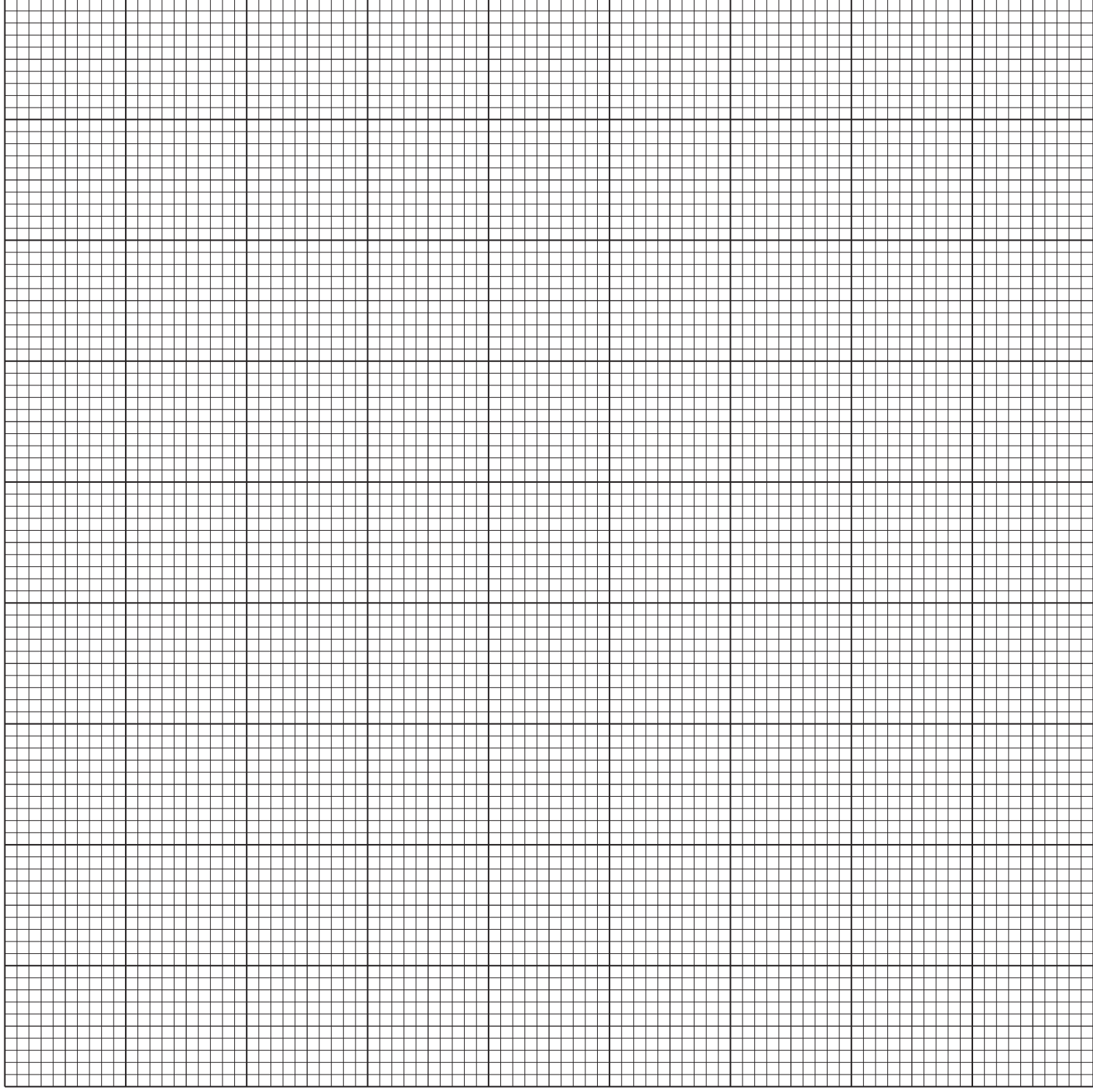
I	
II	
III	
IV	

[4]

- (b) (i)** Plot a graph of the maximum temperature (*y*-axis) against the volume of **FB 5** (*x*-axis) on the grid. The scale on the *y*-axis should be suitable for temperature readings to be 2 °C above the largest maximum temperature.

Label any points you consider to be anomalous.

Draw **two** lines of best fit, the first for the increase in maximum temperature and the second for after the largest maximum temperature has been reached. Extrapolate both lines so that they intersect.



[4]

I		II		III		IV	
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- (ii)** Use the intersection on your graph in **(b)(i)** to determine the volume of **FB 5** required to neutralise 10.0 cm³ of **FB 4**.

volume of **FB 5** = cm³ [1]

(c) Calculations

- (i)** Calculate the amount, in mol, of hydrochloric acid in the volume of **FB 5** in **(b)(ii)**.
(If you were unable to determine an answer to **(b)(ii)**, use 5.10 cm³ as the volume of **FB 5**. This may **not** be the correct answer.)

amount of HCl = mol

Deduce the amount, in mol, of sodium hydroxide in 10.0 cm³ of **FB 4**.

amount of NaOH = mol [1]

- (ii)** Calculate the energy change, in J, when the amounts of reagents in **(c)(i)** neutralise each other. Show your working.

energy change = J [1]

- (iii)** Use your answer to **(c)(ii)** to calculate the enthalpy change, in kJ mol⁻¹, when one mole of **FB 4** is neutralised by one mole of **FB 5**.

enthalpy change = kJ mol⁻¹ [1]
sign value

[Total: 12]

Qualitative analysis

For each test you should record all your observations in the spaces provided.

Examples of observations include:

- colour changes seen
- the formation of any precipitate and its solubility (where appropriate) in an excess of the reagent added
- the formation of any gas and its identification (where appropriate) by a suitable test.

You should record clearly at what stage in a test an observation is made.

Where no change is observed, you should write ‘no change’.

Where reagents are selected for use in a test, the name or correct formula of the element or compound must be given.

If any solution is warmed, a boiling tube must be used. If a solid is heated, a hard-glass test-tube must be used.

Rinse and reuse test-tubes and boiling tubes where possible.

No additional tests should be attempted.